Alex Orri Ingvarsson

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Education

Imperial College London **Sep 2026 – Sep 2027**
MSc Advanced Computing *London, United Kingdom*
- Modules: Advanced Computer Architecture, System Performance Engineering, ML Systems and Hardware

EPFL (École Polytechnique Fédérale de Lausanne) **Sep 2023 – Jul 2026**
BSc Computer Science, grade 5.43/6 *Lausanne, Switzerland*
- Coursework: Computer Systems (OS, networking, C), Systems Programming, Computer Architecture (VHDL)

Experience

Arago **May 2026 – Sep 2026**
Member of Technical Staff (AI accelerator system-on-chip) *Paris, France*
- Owned the end-to-end C++/C software stack for an optical AI accelerator SoC: PyTorch backend, Linux kernel driver, user-space runtime and on-device firmware; offered a full-time role to lead it
- Rebuilt all on-device firmware from scratch for a new multi-core SoC; ported the host-side runtime and driver
- Designed the host–device transport protocol: single-writer command and completion ring buffers
- Cut kernel integration to 1 header declaration with a Python code generator producing the C++ serialisation
- Wrote custom linker scripts for on-chip memory placement and bare-metal per-core tests for RTL verification
- Debugged firmware through RTL simulation traces (Cadence Xcelium) and Arm Fast Models virtual platforms
- Led 3 interns, each owning one layer of the stack; set up CMake/Make builds and GitLab CI pipelines

EPFL Racing Team **Sep 2024 – Jul 2025**
Software Engineer *Lausanne, Switzerland*
- Rebuilt the real-time vehicle control codebase (LabVIEW) driving physical sensors and actuators as modular code
- Implemented structured error handling enabling rapid diagnosis of hardware–software failures during live car testing

Ása Tech **Jul 2024 – Aug 2025**
CTO *Reykjavik, Iceland (Remote)*
- Led development of an AI-powered healthcare triage platform serving patients in Iceland; secured investment

SimplyBook.me **Jul 2022 – Aug 2023**
Fullstack Software Developer *Limassol, Cyprus*
- Shipped production features (React, PHP/Symfony, MariaDB) to a booking platform serving thousands of businesses

Projects

Distributed Key-Value Store | *C, POSIX threads, UDP* **2025**
- Built a Dynamo-style distributed hash table with SHA-1 consistent hashing and a quorum protocol (N/R/W)
- Wrote a multithreaded server (pthreads) and a UDP network layer with request routing, timeouts and error recovery

Real-Time EEG Stimulus Pipeline | *Python, PsychoPy, LSL, FastAPI* – London Neurotech Hackathon **2026**
- Architected a multi-process orchestrator synchronising 4 live streams (EEG, voice, vision, stimulus) to the ms
- Implemented frame-accurate stimulus display (vsync-locked OpenGL and browser backends); ∼5,300 lines in 2 days

Technical Skills

Languages: C++, C, Python, Java, Scala, SQL, TypeScript
Systems & Hardware: bare-metal firmware, device drivers, linker scripts, host–device protocols, RTL simulation, Arm Fast Models, pthreads
Machine Learning: PyTorch (custom device backends, ATen kernels, dispatch and fallbacks), NumPy, scikit-learn
Tools: Linux, Git, GitLab CI, CMake, Make, GDB, Docker, LabVIEW
Spoken languages: English (fluent), French (fluent), Icelandic (native)